Solution Requirements

Date	30 June 2026
Team ID	SWTID-2026-8096
Project Name	Rising Waters
Maximum Marks	4 Marks

Functional and Non-Functional Requirements:

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can access the flood prediction system securely through the web application (no login required for basic prediction).	Medium
2	Authorization levels	General users can enter rainfall data and view prediction results. The administrator can update the ML model and dataset.	Medium
3	External interfaces	The system interacts with the trained Machine Learning model and historical rainfall dataset for prediction.	High

4	Transactions processing	The system accepts rainfall values, validates them, processes the prediction, and displays the result instantly.	High
5	Reporting	Displays **Flood Chance** or **No Flood Chance** along with safety recommendations on the result page.	High
6	Business rules, etc.	Prediction is generated only after all required rainfall fields contain valid values.	High
7	Compliance to laws or regulations	User inputs are processed securely, and no personal information is stored without consent.	Medium
8	*Other*	The application provides a simple and user-friendly interface for flood prediction.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate flood predictions quickly.	Prediction result displayed in less than 3 seconds.
2	Scalability	The application should support multiple users simultaneously without significant delay.	Supports 100+ concurrent users.
3	Security & Data Privacy	User input data should be processed securely, and unauthorized access should be prevented.	Secure data handling using HTTPS and input validation.
4	Reliability & Availability	The system should remain available and provide consistent predictions.	99% uptime during normal operation.
5	Usability & Accessibility	The interface should be simple, responsive, and easy to use on desktop and mobile devices.	Responsive design with intuitive navigation.
6	Other	Maintainability	The system should allow easy updates to the machine learning model and dataset without affecting core functionality

The **Rising Waters – AI-Based Flood Prediction System** is designed to provide accurate flood risk predictions using rainfall data and a trained machine learning model. The functional requirements define the system's core features, including rainfall data input, prediction processing, and result display. The non-functional requirements ensure the application performs efficiently, remains secure and reliable, and delivers a user-friendly experience across different devices. This specification provides a clear foundation for the system's development and deployment.